

The 64-bit OS

To bring the true power of Itanium to light, you need 64-bit operating systems and applications and as of now there are no 64-bit operating systems available.

However, Windows has already announced two operating systems for Intel's Itanium processor, Windows Advanced Server Limited Edition and Windows XP 64-Bit Edition for workstations. Windows XP 64-Bit Edition will not only support technical and scientific applications, but also run 32-bit business and enterprise applications. Future offerings from Microsoft will include server applications including the SQL Server 2000. Hence, with the introduction of its 64-bit OS, Microsoft will be competing with Unix vendors as well as Linux in this market.

dows 98, Me, 2000 and NT. In the next logical iteration, Itanium's 64-bit architecture is built to tap the potential that is going to be offered by 64-bit applications and operating systems, such as the Windows XP 64-Bit scheduled to be launched in October this year.

Until now, processors have been fabricated with two levels of cache (L1 and L2) onboard, but the Itanium has three levels of cache (L1, L2 and L3). It comes with 128 KB of L1 cache, while the L2 cache ranges from 256 KB to 1 MB. The L1 and L2 cache are on-die. OEMs will be able to add L4 cache too. On-die cache, which refers to the high-speed memory that's fabricated onto the processor core itself, is a common feature available in today's processors, be it for desktops or servers. Since it is on-die, cache is run at full speed, thereby processing data at a faster rate than processors that do not have on-die cache.

The L3 cache is mounted separately on the card that houses the actual Itanium core and ranges from 2 MB to 4 MB. The cartridge houses this daughter card—much like what was seen with Intel's slot-based processors. The L3 and L4 cache will not be on-die but will be unified with the package. The additional cache is handy if you are using high-end applications such as data warehousing or running a Web server.

As of now, the Itanium is available in speeds of 733 MHz and 800 MHz and just like the Pentium 4, this processor will also be based upon Intel's Quad Pumped bus technology to deliver an effective bus frequency of 400 MHz, while the actual bus runs at 100 MHz.

On an EPIC path...

Another aspect of the Itanium is the use of enhanced parallel processing techniques, which basically means its ability to process more than one instruction at a time. EPIC, the name given to this technology, executes and processes information in parallel and on an instruction level. In this system, the IA-64 compiler splits the source code into sets of machine code that are fed to the processor in parallel. This increased co-operation between the hardware and compiler sub-systems of the processor greatly increases its efficiency as compared to its predecessors, allowing it to execute more instructions in the same amount of time.

So far the Pentium III and the Athlon could issue three instructions per clock cycle but the Itanium, with a parallel execution core, will have a six issue scheduler, 11 execution units, four integer, two floating point, three branch and two load/store units. This implies that the Itanium can handle an enormous number of instructions simultaneously.

In order to implement EPIC, two technologies are used: 'Predication' and 'Speculation'.

Predication: Today's processors use branch prediction (the path the processor needs to decide on while processing data) but a considerable amount of time is spent on calculations for branches that end up being unused and finally discarded. Predication is a compiler-based technique of looking ahead to make more accurate predictions of what code will be actually used, thereby considerably reducing chances of making wrong predictions and wasting precious clock cycles.

Speculation: This is a process whereby a processor loads instruction sets and data even before you ask for it. The Itanium processor performs this task while it is idle, which means that the relevant data and instructions are already present and available when the processor needs them, thereby increasing its efficiency.

Similar to other processors, the Itanium preloads instruction sets and data purely through a set of algorithms based on probability and there are chances that the instructions loaded might be of no use at all. In this case, the unused instructions and data are simply flushed from the processor and alternate courses of action are taken. The advantage of speculation is evident only when the predictions made by the processor are correct and when the pre-fetched data is actually needed.

Where does it lead?

So what does a 64-bit processor mean in real life? It simply means that the processor is able to access 64 bits of memory

Infographic: Jaya Shetty

